

**ĐẠI HỌC QUỐC GIA HÀ NỘI
TRƯỜNG ĐẠI HỌC CÔNG NGHỆ**



**BÁO CÁO THỰC HÀNH
XỬ LÝ TÍN HIỆU SỐ**

Nguyễn Đức Giang

Bài 1

```
fs0 = 250;

if ~exist('ECG0', 'var')
    t_test = 0:1/fs0:2;
    ECG0 = sin(2*pi*1.2*t_test) +
0.5*sin(2*pi*50*t_test) + ...
        0.2*sin(2*pi*100*t_test) +
0.1*randn(size(t_test));
end

N = length(ECG0);
t = (0:N-1)/fs0;

Y = fft(ECG0);
P2 = abs(Y/N);
P1 = P2(1:floor(N/2)+1);
P1(2:end-1) = 2*P1(2:end-1);
f = fs0*(0:(floor(N/2)))/N;

figure('Color', 'w');

subplot(2,1,1);
plot(t, ECG0, 'b');
title('ECG0 - Time Domain');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(2,1,2);
plot(f, P1, 'k'); hold on;
title('ECG0 - Amplitude Spectrum');
xlabel('Frequency (Hz)');
ylabel('|P1(f)|');
grid on;
axis([0 125 0 max(P1)*1.1]);

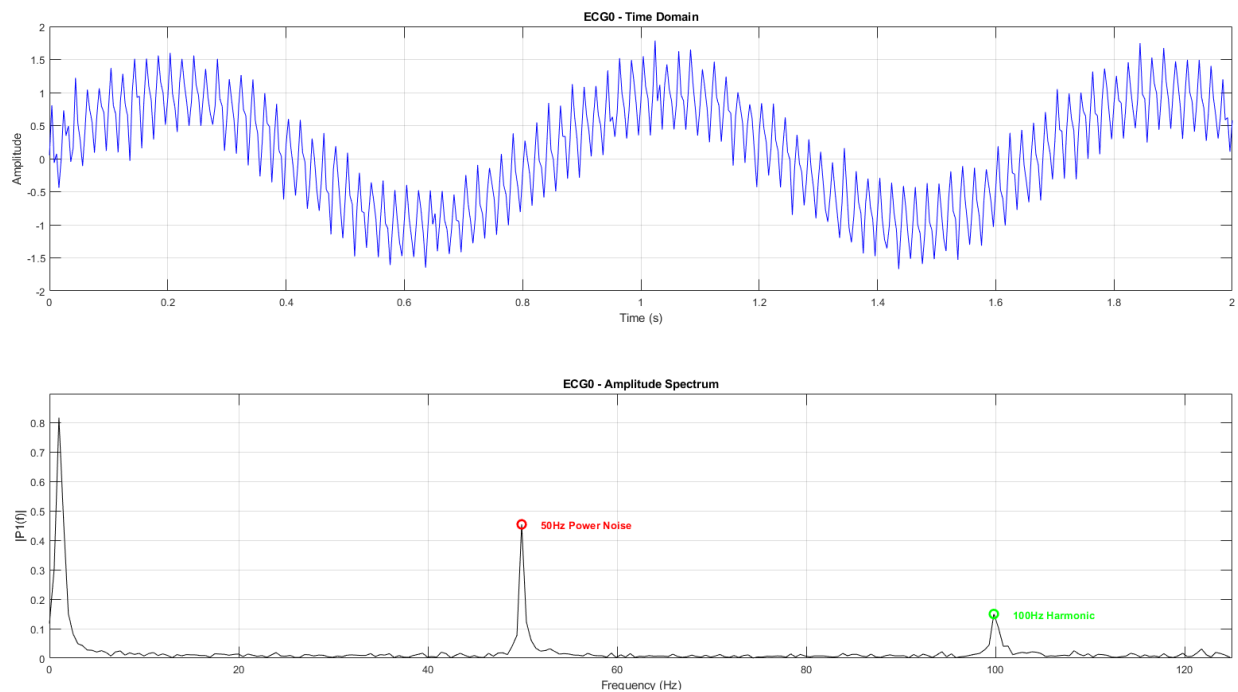
[~, idx50] = min(abs(f - 50));
[~, idx100] = min(abs(f - 100));
```

```

plot(f(idx50), P1(idx50), 'ro', 'MarkerSize', 8,
'LineWidth', 2);
text(f(idx50)+2, P1(idx50), '50Hz Power Noise',
'Color', 'r', 'FontWeight', 'bold');

plot(f(idx100), P1(idx100), 'go', 'MarkerSize', 8,
'LineWidth', 2);
text(f(idx100)+2, P1(idx100), '100Hz Harmonic',
'Color', 'g', 'FontWeight', 'bold');

```



Bài 2

```

fs0 = 250;

if ~exist('ECG0', 'var')
    t_test = 0:1/fs0:2;
    ECG0 = sin(2*pi(1.2)*t_test) +
0.6*sin(2*pi*50*t_test) + ...
        0.3*sin(2*pi*100*t_test) +
0.1*randn(size(t_test));

```

```

end

wo = 50/(fs0/2);
bw = wo/35;
[b_notch, a_notch] = iirnotch(wo, bw);

[b_lp, a_lp] = butter(4, 100/(fs0/2), 'low');

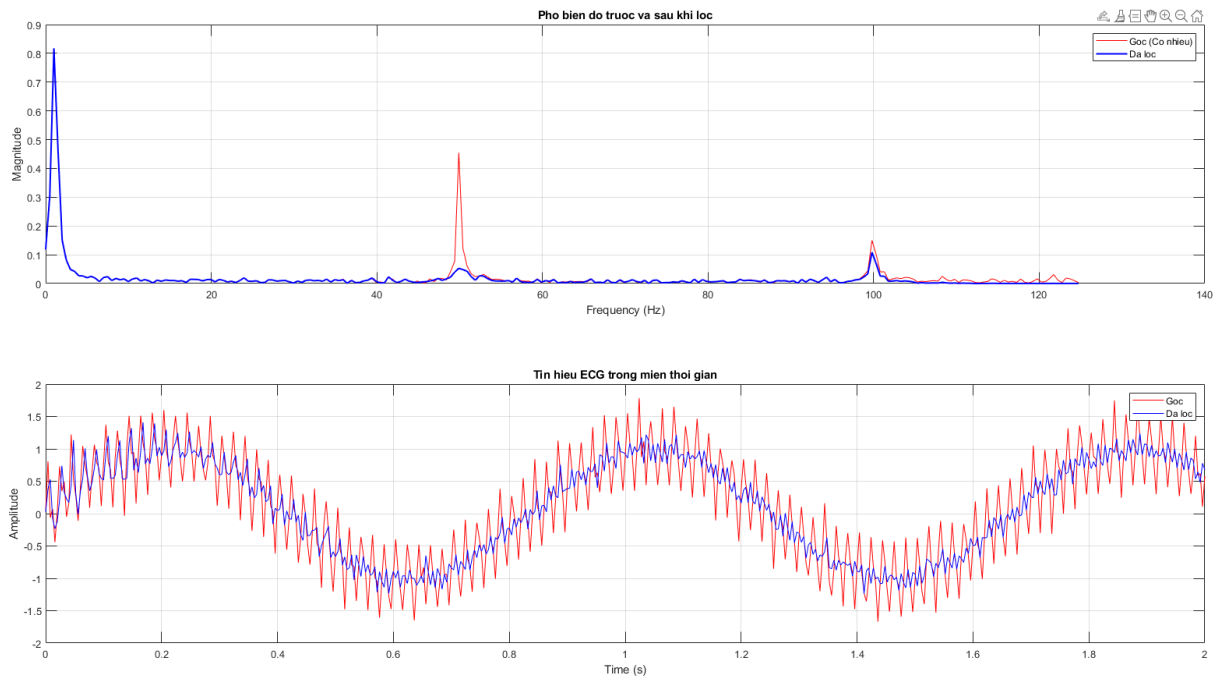
ECG_filtered = filter(b_notch, a_notch, ECG0);
ECG_filtered = filter(b_lp, a_lp, ECG_filtered);

N = length(ECG0);
f = fs0*(0:(floor(N/2)))/N;
P_orig = abs(fft(ECG0)/N); P_orig =
P_orig(1:floor(N/2)+1); P_orig(2:end-1) =
2*P_orig(2:end-1);
P_filt = abs(fft(ECG_filtered)/N); P_filt =
P_filt(1:floor(N/2)+1); P_filt(2:end-1) =
2*P_filt(2:end-1);

figure('Color', 'w');
subplot(2,1,1);
plot(f, P_orig, 'r'); hold on;
plot(f, P_filt, 'b', 'LineWidth', 1.5);
title('Pho bien do truoc va sau khi loc');
xlabel('Frequency (Hz)'); ylabel('Magnitude');
legend('Goc (Co nhieu)', 'Da loc');
grid on;

subplot(2,1,2);
plot((0:N-1)/fs0, ECG0, 'r'); hold on;
plot((0:N-1)/fs0, ECG_filtered, 'b');
title('Tin hieu ECG trong mien thoi gian');
xlabel('Time (s)'); ylabel('Amplitude');
legend('Goc', 'Da loc');
grid on;

```



Bài 3

```
fs0 = 250;

if ~exist('ECG0', 'var')
    t_test = 0:1/fs0:2;
    ECG0 = sin(2*pi*1.2*t_test) + 0.6*sin(2*pi*50*t_test) + ...
        0.3*sin(2*pi*100*t_test) + 0.1*randn(size(t_test));
end

wo = 50/(fs0/2);
bw = wo/35;
[b_notch, a_notch] = iirnotch(wo, bw);

[b_lp, a_lp] = butter(4, 100/(fs0/2), 'low');

fECG0 = filter(b_notch, a_notch, ECG0);
fECG0 = filter(b_lp, a_lp, fECG0);

N = length(fECG0);
t = (0:N-1)/fs0;
```

```

Y_f = fft(fECG0);
P2_f = abs(Y_f/N);
P1_f = P2_f(1:floor(N/2)+1);
P1_f(2:end-1) = 2*P1_f(2:end-1);
f = fs0*(0:(floor(N/2)))/N;

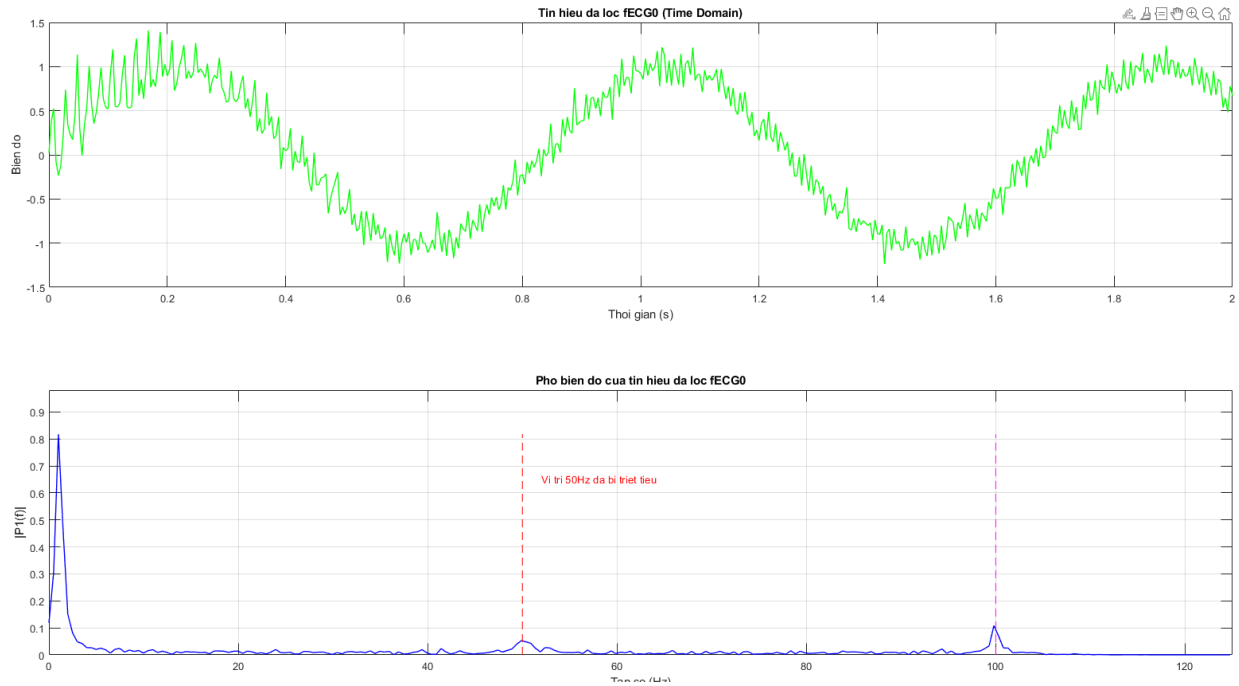
figure('Color', 'w');

subplot(2,1,1);
plot(t, fECG0, 'g', 'LineWidth', 1);
title('Tin hieu da loc fECG0 (Time Domain)');
xlabel('Thoi gian (s)');
ylabel('Bien do');
grid on;

subplot(2,1,2);
plot(f, P1_f, 'b', 'LineWidth', 1);
title('Pho bien do cua tin hieu da loc fECG0');
xlabel('Tan so (Hz)');
ylabel('| P1(f) |');
grid on;
axis([0 125 0 max(P1_f)*1.2]);

[~, idx50] = min(abs(f - 50));
[~, idx100] = min(abs(f - 100));
line([50 50], [0 max(P1_f)], 'Color', 'r', 'LineStyle', '--');
line([100 100], [0 max(P1_f)], 'Color', 'm', 'LineStyle', '--');
text(52, max(P1_f)*0.8, 'Vi tri 50Hz da bi triet tieu', 'Color', 'r');

```



Bài 4

```
fs0 = 250;

% 1. Tao/Nap du lieu ECG0
if ~exist('ECG0', 'var')
    t_test = 0:1/fs0:2;
    ECG0 = sin(2*pi*1.2*t_test) + 0.5*sin(2*pi*50*t_test) + ...
        0.2*sin(2*pi*100*t_test) + 0.1*randn(size(t_test));
end

% 2. Thiet ke bo loc va tao fECG0 (da loc)
wo = 50/(fs0/2);
bw = wo/35;
[b_notch, a_notch] = iirnotch(wo, bw);
[b_lp, a_lp] = butter(4, 100/(fs0/2), 'low');

fECG0 = filter(b_notch, a_notch, ECG0);
fECG0 = filter(b_lp, a_lp, fECG0);

% 3. Tao/Nap du lieu ECG2 (Gia su ECG2 la tin hieu sach de doi chieu)
if ~exist('ECG2', 'var')
```

```

    ECG2 = sin(2*pi*1.2*t_test) + 0.05*randn(size(t_test));
end

% 4. Tinh toan su khac biet
diff_0_2 = ECG0 - ECG2;
diff_f0_2 = fECG0 - ECG2;

% 5. Ve do thi
N = length(ECG2);
t = (0:N-1)/fs0;

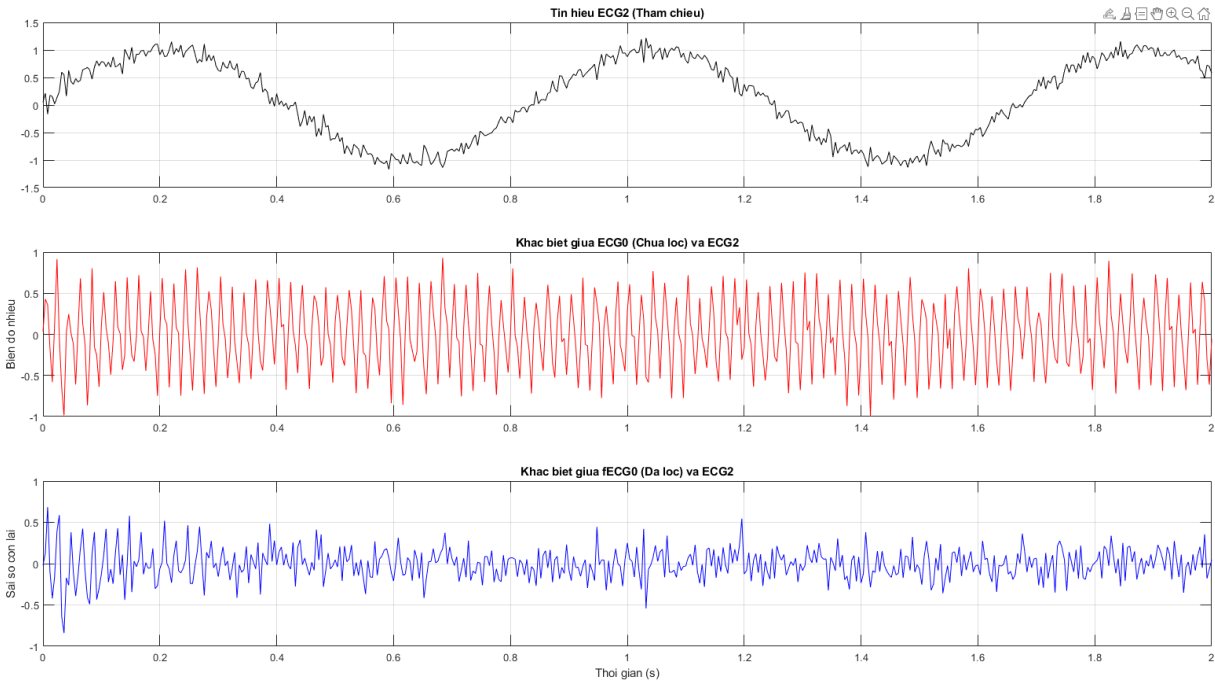
figure('Color', 'w');

subplot(3,1,1);
plot(t, ECG2, 'k');
title('Tin hieu ECG2 (Tham chieu)');
grid on;

subplot(3,1,2);
plot(t, diff_0_2, 'r');
title('Khac biet giua ECG0 (Chua loc) va ECG2');
ylabel('Bien do nhieu');
grid on;

subplot(3,1,3);
plot(t, diff_f0_2, 'b');
title('Khac biet giua fECG0 (Da loc) va ECG2');
xlabel('Thoi gian (s)');
ylabel('Sai so con lai');
grid on;

```



Bài 5

```
function [MSD, MAD, SNR] =
evaluate_filter(x, y)

    x = x - mean(x);
    y = y - mean(y);

    [c, lags] = xcorr(x,
y);
    [~, idx] = max(c);
    delay = lags(idx);

    if delay > 0
        x_sync = x(1 :
end-delay);
        y_sync = y(delay+1
: end);
    elseif delay < 0
        x_sync =
x(abs(delay)+1 : end);
        y_sync = y(1 :
end-abs(delay));
```

```
[msd_val, mad_val,
snr_val] =
evaluate_filter(ECG2,
fECG0);
figure('Color', 'w');
plot(ECG2, 'k'); hold on;
plot(fECG0, 'r--');
legend('Tham chieu
(ECG2)', 'Da loc
(fECG0)'); title('So sanh
tin hieu truoc khi bu
tre');
```

```

else
    x_sync = x;
    y_sync = y;
end

L = length(x_sync);
diff = x_sync -
y_sync;

MSD = sum(diff.^2) /
L;

MAD = sum(abs(diff)) /
L;

Px = sum(x_sync.^2);
SNR = Px /
sum(diff.^2);

SNR_dB = 10 *
log10(SNR);

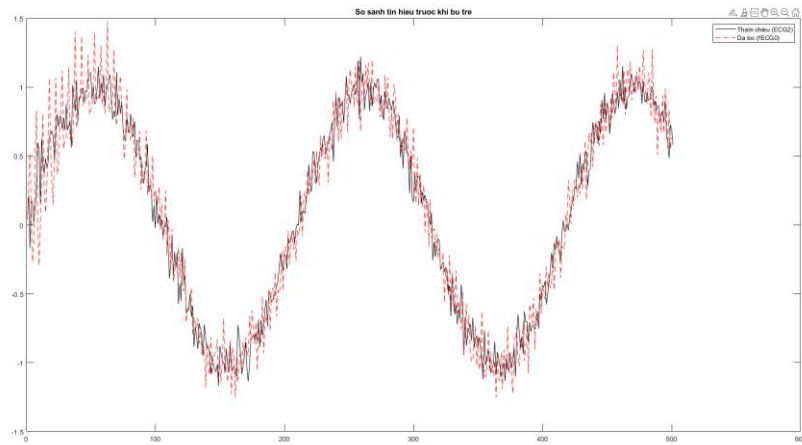
fprintf('--- Ket qua
danh gia bo loc ---\n');
fprintf('Khoang tre
tinh toan: %d mau\n',
abs(delay));
fprintf('MSD: %.6f\n',
MSD);
fprintf('MAD: %.6f\n',
MAD);
fprintf('SNR: %.2f
(tuyen tinh) hoac %.2f
dB\n', SNR, SNR_dB);
end

```

```

--- Ket qua danh gia bo loc ---
Khoang tre tinh toan: 1 mau
MSD: 0.036519
MAD: 0.147294
SNR: 14.17 (tuyen tinh) hoac 11.51 dB
>> evaluate filter

```



Bài 6

```

fs = 250;

if ~exist('ECG0', 'var')
    t_test = 0:1/fs:2;
    ECG0 = sin(2*pi*1.2*t_test) +
0.5*sin(2*pi*50*t_test) + 0.2*sin(2*pi*100*t_test) +
0.1*randn(size(t_test));
end

if ~exist('ECG1', 'var')
    ECG1 = ECG0 + 0.05*randn(size(ECG0));
end

if ~exist('ECG2', 'var')
    t_test = 0:1/fs:2;
    ECG2 = sin(2*pi*1.2*t_test) +
0.02*randn(size(t_test));
end

wo = 50/(fs/2);
bw = wo/35;

```

```

[b_n, a_n] = iirnotch(wo, bw);
[b_l, a_l] = butter(4, 100/(fs/2), 'low');

fECG0 = filter(b_n, a_n, ECG0);
fECG0 = filter(b_l, a_l, fECG0);

x = ECG2 - mean(ECG2);
y = fECG0 - mean(fECG0);

[c, lags] = xcorr(x, y);
[~, idx] = max(c);
delay = lags(idx);

if delay > 0
    x_s = x(1 : end-delay);
    y_s = y(delay+1 : end);
elseif delay < 0
    x_s = x(abs(delay)+1 : end);
    y_s = y(1 : end-abs(delay));
else
    x_s = x;
    y_s = y;
end

L = length(x_s);
d = x_s - y_s;

MSD = sum(d.^2) / L;
MAD = sum(abs(d)) / L;
SNR = sum(x_s.^2) / sum(d.^2);

fprintf('MSD: %.6f\nMAD: %.6f\nSNR: %.2f\n', MSD, MAD,
SNR);

[qrs_i1, qrs_c1] = pan_tompkin(ECG1, fs, 0);
[qrs_i2, qrs_c2] = pan_tompkin(ECG2, fs, 0);

figure('Color', 'w');
subplot(2,1,1);
plot(ECG1); hold on;

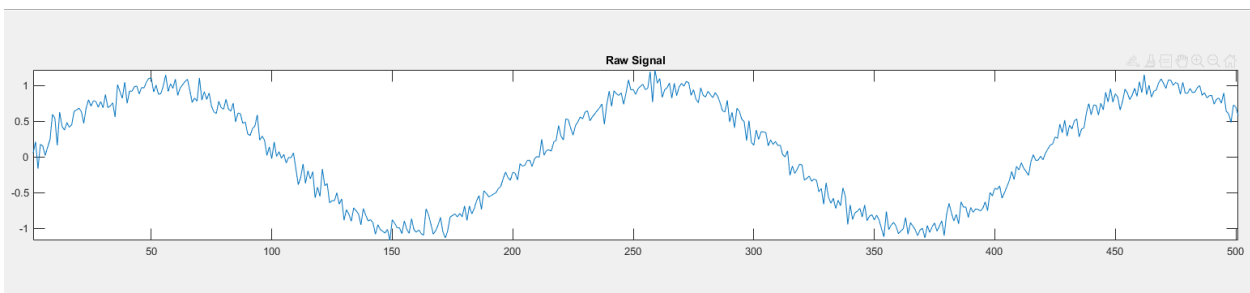
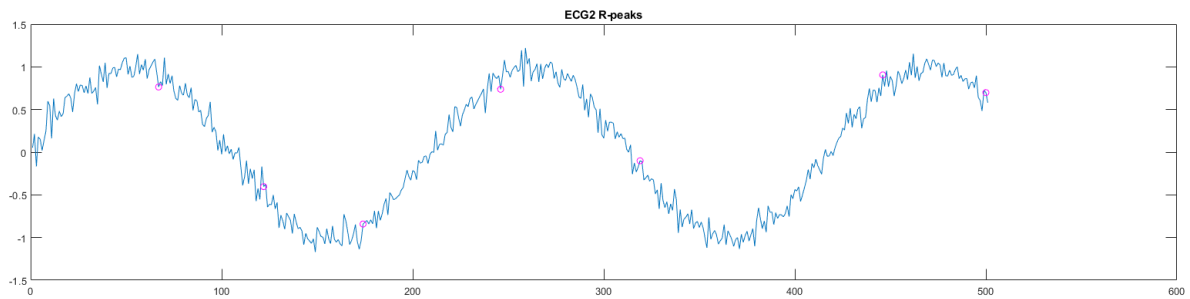
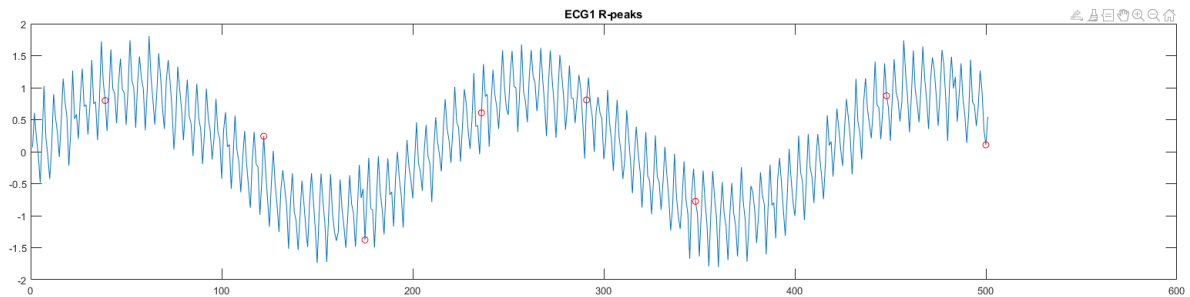
```

```

plot(qrs_i1, ECG1(qrs_i1), 'ro');
title('ECG1 R-peaks');

subplot(2,1,2);
plot(ECG2); hold on;
plot(qrs_i2, ECG2(qrs_i2), 'mo');
title('ECG2 R-peaks');

```



Bài 7

```

diff_i1 = diff(qrs_i1);
mean_rr_sample1 = mean(diff_i1);
mean_rr_sec1 = mean_rr_sample1 / fs;
bpm1 = 60 / mean_rr_sec1;

```

```

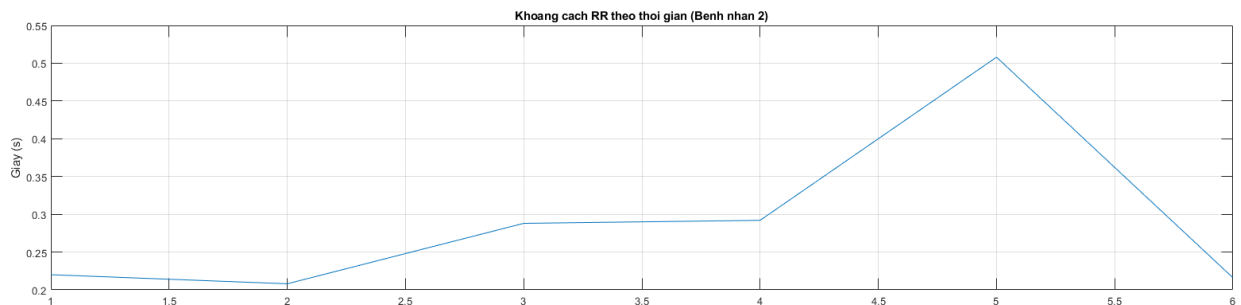
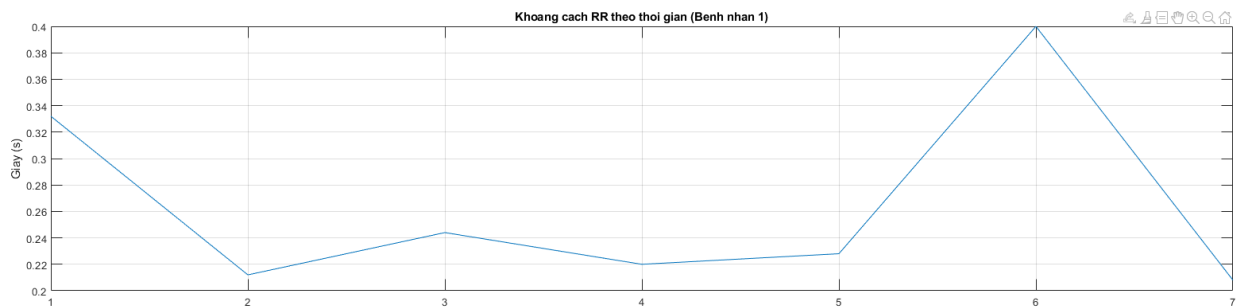
diff_i2 = diff(qrs_i2);
mean_rr_sample2 = mean(diff_i2);
mean_rr_sec2 = mean_rr_sample2 / fs;
bpm2 = 60 / mean_rr_sec2;

fprintf('Nhịp tim trung bình bệnh nhân 1: %.2f BPM\n',
bpm1);
fprintf('Nhịp tim trung bình bệnh nhân 2: %.2f BPM\n',
bpm2);

figure('Color', 'w');
subplot(2,1,1);
plot(diff_i1 ./ fs);
title('Khoảng cách RR theo thời gian (Bệnh nhân 1)');
ylabel('Giây (s)');
grid on;

subplot(2,1,2);
plot(diff_i2 ./ fs);
title('Khoảng cách RR theo thời gian (Bệnh nhân 2)');
ylabel('Giây (s)');
grid on;

```



Bài 8

```
if ~exist('fs', 'var'), fs = 250; end
if ~exist('t1', 'var'), t1 = (0:length(ECG1)-1)/fs;
end

rr_intervals = diff(qrs_i1);
mean_rr = mean(rr_intervals);

lower_bound = 0.5 * mean_rr;
upper_bound = 1.5 * mean_rr;

abnormal_idx = find(rr_intervals < lower_bound |
rr_intervals > upper_bound);

r_abnormal_indices = [];
if ~isempty(abnormal_idx)
    for i = 1:length(abnormal_idx)
        idx = abnormal_idx(i);
        r_abnormal_indices = [r_abnormal_indices,
qrs_i1(idx), qrs_i1(idx+1)];
    end
end
r_abnormal_indices = unique(r_abnormal_indices);

figure('Color', 'w');
plot(t1, ECG1, 'b'); hold on;
plot(t1(qrs_i1), ECG1(qrs_i1), 'go', 'MarkerSize', 4);

if ~isempty(r_abnormal_indices)
    plot(t1(r_abnormal_indices),
ECG1(r_abnormal_indices), 'ro', 'MarkerSize', 8,
'LineWidth', 2);
end

title('Phat hien nhip tim roi loan tren ECG1');
xlabel('Thoi gian (s)');
ylabel('Bien do');
legend('ECG1', 'Dinh R binh thuong', 'Dinh R bat
thuong');
grid on;
```

